



Figure 5: Installation in progress

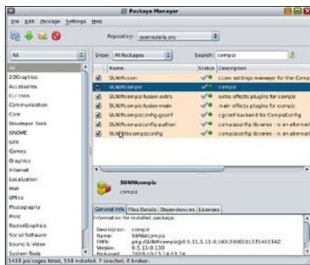


Figure 6: OpenSolaris Package Manager

menu file. Edit this file and add the entries from your Linux *menu.lst*.

Root shell: The first user to be created has super user privileges. That user can prefix any command with *pfexec* to run it with super user privileges (similar to *sudo* the command Linux users are familiar with). For example, to get a root shell, run *pfexec bash*.

You are now all set to start exploring your new OpenSolaris installation! The first thing is the customised default artwork—I don't know what others think, but I find it quite attractive.

OpenSolaris 2008.11 uses GNOME 2.24 as the desktop and introduces a whole bunch of improvements, including an improved package manager, automatic backup management, and a desktop search tool, among others.

More software

The package manager can be used to install software from a collection of network repositories (Figure 6). Software available on the official OpenSolaris repositories

include OpenOffice.org, Netbeans, Eclipse, Songbird, Gobby and the optimised OpenSolaris Web Stack including frameworks like Drupal and Django, several of them instrumented with DTrace probes allowing you to dynamically trace your Web application for improved performance. (You can get more details about the Web Stack at tinyurl.com/sun-web-stack.)

You can launch the package manager from the toolbar or *System*→*Administration*→*Package Manager*. The interface should be familiar to anyone who has used Synaptic or one of the other GUI package managers in Linux.

Note: If you prefer the command line, run *pkg help* in your terminal window to get details of the command line interface.

Some encumbered software that cannot be freely redistributed because of licensing constraints are available in the OpenSolaris Extras repository. Access to this repository requires you to sign up for a Sun Online account. Point your browser to pkg.sun.com and follow the instructions to sign up and log in. You will be redirected to the 'Certificate Requests' page where you can request access to the OpenSolaris Extras repository. Follow the instructions on the page to set up access to the repository. (You will need to run a bunch of commands on the command line. The package manager GUI does not support this currently. However, once you have completed the one-time task of configuring access to the Extras repository, you can use the GUI to browse and install applications.)

Once you have the Extras repository configured, you can install packages from there—for instance, VirtualBox or the Flash plug-in for Firefox. Make sure you choose the right repository in the top panel of the package manager before you start looking for these packages.

The package manager now also includes an update manager that notifies the user when there are updates available.

What's in a filesystem?

OpenSolaris uses ZFS as the default filesystem. ZFS is a new kind of filesystem that provides simple administration, transactional semantics, end-to-end data integrity, and immense scalability. Unlike conventional filesystems, ZFS has integrated volume management features that make it much easier to perform traditionally complicated tasks like resizing filesystems, adding new disks, etc. For example, adding a new disk is as simple as adding the device to the ZFS storage pool (*zpool add*). The extra space is instantly available to all filesystems in the pool.

ZFS also has the ability to create instantaneous snapshots and clones of your filesystems. Snapshots and clones occupy no additional disk space in the pool when initially created, and any unchanged data is shared among the filesystem and all its snapshots and clones.